

Environment and how to run

A single file, `lecture6_exercise.py` — **Python standard library only**, no GPU

Download `lecture6_exercise.py` from the course repository

Implement each function marked `TODO`

Run the file to self-check against the built-in tests

```
python lecture6_exercise.py
```

github.com/mpkato/ir_lecture_slides → `lecture6_exercise.py`

The corpus and qrels

A 15-document corpus, already tokenized. Three queries come with **graded relevance** (2 / 1 / 0):

```
CORPUS = {
    "d01": "Kyoto foliage is the "
           "highlight of autumn . ...",
    ...
}
QUERIES = {"q1": "Kyoto foliage", ...}
QRELS = {
    "q1": {"d01": 2, "d08": 2,
           "d04": 1, "d07": 1, "d03": 0},
    ...
}
```

Part	Task	Points
Core	Task 1 — Inverted index	15
Core	Task 2 — Boolean retrieval (AND / OR)	10
Core	Task 3 — TF-IDF + cosine similarity	15
Core	Task 4 — BM25	15
Core	Task 5 — Evaluation metrics (P@k, AP, nDCG)	15
Advanced	Q1 / Q2 / Q3 — analysis and discussion	10 each (30)
	Total	100

Core tasks are checked by the built-in tests. Advanced tasks are graded on the correctness of your results and the quality of your discussion

Deliverables and naming

Two files, each named **starting with your student number**:

1. Your completed `lecture6_exercise.py` (core tasks)
2. A **PDF** report for the Advanced tasks (Q1–Q3)

```
20XXXXXXX_lecture6_exercise.py  
20XXXXXXX_advanced.pdf
```

Where and when

Upload both files to the course **Dropbox file request**:

<https://www.dropbox.com/request/iknk74uziz8s7vgh1alj>

Before the second lecture from today begins

Note on generative AI — Core tasks are checked by **hidden tests** beyond those shown, on a corpus that may differ; hard-coding the visible outputs will not pass. Advanced tasks are graded on *your* experimental results — write the discussion in your own words, citing the numbers your own implementation produced